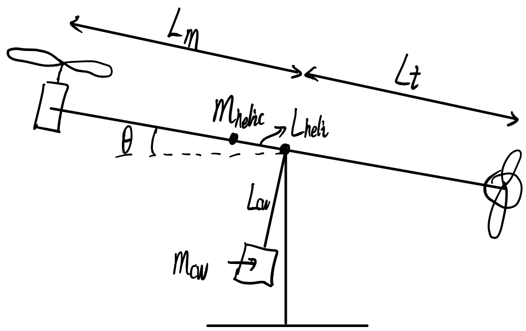
Lagrange's equation

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_k} - \frac{\partial L}{\partial q_k} = \tau_k \qquad L = T - V$$



$$q = \begin{bmatrix} \theta \\ \psi \end{bmatrix}$$

$$T = \frac{1}{2} w^T J w$$

$$T = \frac{1}{2}\int v^2\, dm \qquad v = w \times r \qquad (a \times b)\cdot c = a\cdot(b \times c)$$

$$T = \frac{1}{2}\int (w \times r)\cdot(w \times r)\,dm = \frac{1}{2}\int w \cdot [r \times (w \times r)]\,dm$$

$$T = \frac{1}{2} w^T J w$$

$$T = \frac{1}{2}\begin{bmatrix} q & r \end{bmatrix} \cdot \begin{bmatrix} I_{yy} & -I_{yz} \\ -I_{zy} & I_{zz} \end{bmatrix}\begin{bmatrix} q \\ r \end{bmatrix}$$ symmetric in y direction.

$$T = \frac{1}{2}\begin{bmatrix} q & r \end{bmatrix} \cdot \begin{bmatrix} I_{yy} & 0 \\ 0 & I_{zz} \end{bmatrix}\begin{bmatrix} q \\ r \end{bmatrix}$$

$$T = \frac{1}{2}\begin{bmatrix} q & r \end{bmatrix} \cdot \begin{bmatrix} I_{yy}\cdot q \\ I_{zz}\cdot r \end{bmatrix} = \frac{1}{2} I_{yy} q^2 + \frac{1}{2} I_{zz} r^2$$

$$\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi \\ \frac{\sin\phi}{\cos\theta} & \frac{\cos\phi}{\cos\theta} \end{bmatrix}\begin{bmatrix} q \\ r \end{bmatrix} \qquad \phi = 0$$

$$\begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\cos\theta} \end{bmatrix}\begin{bmatrix} q \\ r \end{bmatrix} \qquad \left.\begin{aligned} \dot{\theta} = q &\rightarrow q = \dot{\theta} \\ \dot{\psi} = \frac{r}{\cos\theta} &\rightarrow r = \cos\theta\,\dot{\psi} \end{aligned}\right\}$$

$$T = \frac{1}{2} I_{yy}\dot{\theta}^2 + \frac{1}{2} I_{zz}\cos^2\theta\,\dot{\psi}^2$$

$$V(\theta) = m_{heli} g L_{heli} \cdot \sin\theta + m_{cw} g L_{cw} \cdot (1-\cos\theta)$$

$$L = T - V = \frac{1}{2} I_{yy} \dot{\theta}^2 + \frac{1}{2} I_{zz} \cos^2\theta \dot{\psi}^2 - m_{heli} g L_{heli} \cdot \sin\theta - m_{cw} g L_{cw} \cdot (1-\cos\theta)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_k} - \frac{\partial L}{\partial q_k} = \tau_k$$

$$q_1 = \theta : \frac{\partial L}{\partial \dot{\theta}} = I_{yy} \dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = I_{yy} \ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = -I_{zz} \cos\theta \sin\theta \dot{\psi}^2 - m_{heli} g L_{heli} \cdot \cos\theta - m_{cw} g L_{cw} \sin\theta$$

$$I_{yy} \ddot{\theta} + I_{zz} \cos\theta \sin\theta \dot{\psi}^2 + m_{heli} g L_{heli} \cdot \cos\theta + m_{cw} g L_{cw} \sin\theta = \tau_\theta$$

$$q_2 = \psi : \frac{\partial L}{\partial \dot{\psi}} = I_{zz} \cos^2\theta \dot{\psi}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\psi}}\right) = I_{zz} \cos^2\theta \ddot{\psi} - 2 I_{zz} \cos\theta \sin\theta \cdot \dot{\theta} \dot{\psi}$$

$$\frac{\partial L}{\partial \psi} = 0$$

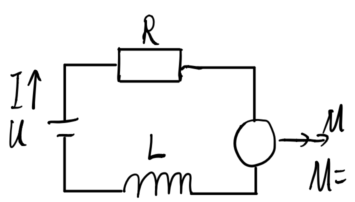
$$I_{zz} \cos^2\theta \ddot{\psi} - 2 I_{zz} \cos\theta \sin\theta \cdot \dot{\theta} \dot{\psi} = \tau_\psi$$

$$\begin{bmatrix} I_{yy} & 0 \\ 0 & I_{zz}\cos^2\theta \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + \begin{bmatrix} 0 & I_{zz} \dot{\psi} \sin\theta \cos\theta \\ -I_{zz} \dot{\psi} \sin\theta\cos\theta & -I_{zz} \dot{\theta} \sin\theta\cos\theta \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} m_{heli} g L_{heli} \cdot \cos\theta + m_{cw} g L_{cw} \sin\theta \\ 0 \end{bmatrix} = \begin{bmatrix} \tau_\theta \\ \tau_\psi \end{bmatrix}$$

Can be used for nonlinear control.

$$\tau_\theta = T_m \cdot L_m$$

$$\tau_\psi = T_t \cdot L_t \cdot \cos\theta$$



$$U = IR + L\frac{dI}{dt} + K_m \cdot W_m \qquad V = L \cdot \frac{dI}{dt} = LsI$$

$$y = K_m \cdot I$$

$$M = K_m \cdot I \qquad \frac{dI}{dt} = -\frac{R}{L} I - \frac{K_m}{L} W_m + \frac{1}{L} u$$

$$y = K_m \cdot I \qquad J_m \dot{w}_m + B_m W_m = K_m \cdot I$$

motor dynamics : $$\begin{bmatrix} \dot{I} \\ \dot{w}_m \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{K_m}{L} \\ \frac{K_m}{J_m} & -\frac{B_m}{J_m} \end{bmatrix} \begin{bmatrix} I \\ w \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} u$$

$$M = [K_m \quad 0] \begin{bmatrix} I \\ \end{bmatrix} + [0] u$$

$$\begin{bmatrix} w \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

$$V_{eft} = u - k_m w_m$$

$$V_{eft} = L_a \frac{dI}{dt} + R_a I \qquad G = \frac{I(s)}{V_{eft}(s)} = \frac{1}{L_a s + R}$$